TECHNICAL DESCRIPTION PAPER

RESCUE MAZE

PRIMEX FCoEE

Faizuddin Centre of Educational Excellence (FCoEE)

Malaysia

Team Name	:	PRIMEX FCoEE
League	:	Rescue Maze
Submission Year	:	2026
Members Name	:	1. Afif (Leader)
		2. Hafiy
		3. Hazieq
		4. Adam
		5. Muqri (Mentor 1)
		6. Ameerul (Mentor 2)

1. INTRODUCTION

Abstract

PRIMEX FCoEE is a fully autonomous maze-solving rescue robot developed by a student team from the Faizuddin Centre of Educational Excellence (FCoEE) for the RoboCupJunior Rescue Maze 2026 competition. The robot is built on a dual-processor architecture, with a Jetson Orin Nano handling all vision and navigation decisions and an ESP32-S3 managing real-time motor and actuator control, mounted on a fully custom chassis 3D-printed using engineering-grade carbon fibre reinforced materials (ABS-CF and PETG-CF) for structural rigidity and durability under field conditions. The robot explores unknown mazes autonomously using a Right-Hand Rule algorithm with real-time 2D grid mapping, requiring no pre-mapping and no hardcoded field logic. It detects two types of victims. The first victim is Greek letter victims (Φ, Ψ, Ω) using a custom TensorFlow deep learning model trained on a manually captured dataset covering varied angles, distances, and lighting, achieving 90% accuracy. The second victim is cognitive targets, detected using outermost-circle-first template strategy that anchors inner ring boundaries and computes the health status sum from colour values, achieving 80% ring layer detection and 90% per-ring colour classification. Floor tile types are reliably identified by a downward-facing mBot2 colour sensor with local calibration performed before each run, allowing the robot to adapt to changing venue lighting conditions.

The full system integrates high-performance hardware, autonomous maze navigation, trained letter victim detection, template-based cognitive target detection, and adaptive floor colour sensing into a single cohesive whole. The combination reflects the team's commitment to building a robot that is not only technically capable across every scoring dimension of the competition, but also robust and consistent under real field conditions.

Team

- **AFIF – Hardware and Movement Lead**
    - Led mechanical design and chassis fabrication. Developed encoder-based motor synchronisation and ESP32 low-level firmware for motor drive, serial communication, and actuator control.
- **HAFIY – Mapping and Navigation Lead**
    - Developed the autonomous maze navigation algorithm using Right-Hand Rule and 2D grid mapping. Integrated ToF sensor data with decision logic on the Jetson Orin Nano and handled checkpoint recovery.
- **HAZIEQ – Image Processing (Greek Letter Detection)**
    - Collected and curated the training dataset for letter victim detection. Trained the TensorFlow CNN model for Φ/Ψ/Ω classification, developed the live inference pipeline, and integrated the rescue kit deployment trigger.
- **ADAM – Image Processing (Cognitive Target Detection)**
    - Developed the concentric ring detection algorithm using OpenCV. Implemented the outermost-circle template strategy, per-ring colour classification using NumPy, and health status computation with the kit drop command.
- **MUQRI – Mentor 1 (Design and Hardware)**
    - Provided consultation on robot mechanical design, component selection, power system architecture, and chassis material choices.
- **AMEERUL – Mentor 2 (Algorithm and Strategy)**
    - Provided consultation on mapping algorithm design, image processing strategy for cognitive targets, and overall competition strategy.

Keywords

Autonomous maze exploration, Right-Hand Rule navigation, convolutional neural network, cognitive target detection, concentric ring template, TensorFlow inference, OpenCV image processing, encoder odometry, carbon fibre 3D printing, dual-processor architecture.

2. PROJECT PLANNING

Teams Objective

- The primary objective of PRIMEX FCoEE is to build a fully autonomous robot capable of exploring an unknown maze, correctly identifying and classifying victims (letter and cognitive targets), deploying rescue kits accurately, and maximising competition score within the 8-minute game window. The secondary objective is to produce high-quality technical documentation that communicates the team's design decisions and innovations clearly.

Requirements Definition

- Requirements were derived from three constraint sources: competition rules, physical limitations of the arena, and team resource constraints.

Requirement	Constraint Source	Target
Robot height ≤ 250 mm	Rules 4.2	200 mm achieved
Robot width fits 280 mm pathway	Rules 3.3	190 mm achieved
Fully autonomous operation	Rules 4.1	No remote control, no pre-mapping
Single start button only	Rules 4.2	Momentary push button on ESP32
Emergency stop button	Rules 4.2 / Safety	Self-latch cut-off button
RGB LED victim indicator (500 ms blink)	Rules 4.2 & 5.6	RGB LED Indicator
Carry handle for robot pick-up	Rules 4.2	3D printed main handle mounted on top body
Max 8 rescue kits	Rules 3.7	8 × 11 mm PLA cubes loaded

Rescue kit ≥ 1 cm^3 volume	Rules 3.7	11×11×11 mm = 1.331 cm^3
Checkpoint map saves (non-volatile)	Rules 3.2 & 5.5	Map saved to Jetson storage on silver tile
Avoid black tiles	Rules 5.5	Colour sensor detects, serial command to stop/reverse
Stop 5s on blue tiles	Rules 3.2 & 5.5	Colour sensor triggers 5-second hold command
Victim detection within 15 cm	Rules 5.6	ToF front-centre triggers stop and RGB LED blink
Maximum voltage 48V DC	Rules 1.3.2	20V battery used

Project Schedule and Milestones

- The project ran from January 2026 to June 2026 (approximately 24 weeks). Tasks were distributed between P1/P2 (hardware, movement, mapping) and P3/P4 (vision and image processing), with regular integration reviews every two weeks.

PRIMEX FCoEE GANTT CHART (SCHEDULE AND MILESTONES)

Phase	Activities and Milestone	JANUARY				FEBRUARY				MARCH				APRIL				MAY				JUNE				JULY	PIC	Gate/Review
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1		
Phase 1: Study & Design	Rules analysis, requirements definition, component survey																										All	Team agreement on requirements list
Phase 1: Study & Design	Conceptual robot design finalised																										P1, P2, M1	Design review with mentors
Phase 2: Hardware Build	3D printing chassis (PETG-CF frame, ABS-CF body)																										P1	Physical fit check, robot width < 280 mm
Phase 2: Hardware Build	Electronics assembly: ESP32, L298N, motors, encoders																										P1, P2	Motor spin test, encoder count verified
Phase 2: Hardware Build	Jetson Orin Nano, ToF sensors, cameras integrated																										P2, P3	All 4 ToF sensors reading simultaneously
Phase 3: Software Low Level	ESP32 firmware: motor sync, colour sensor, servo drop																										P1	Motor sync error < 1 mm confirmed
Phase 3: Software High Level	Jetson: ToF reading, serial comms, basic movement																										P2	Robot moves forward and turns on command
Phase 4: Navigation	Right-Hand Rule and 2D grid mapping implemented																										P2	60% maze traversal success on test arena
Phase 4: Vision	Greek letter dataset collection, CNN training																										P3	90% accuracy on validation set
Phase 4: Vision	Cognitive target ring detection pipeline																										P4	80% ring detection, 90% colour classification
Phase 5: Integration	Full system integration, field testing, LoP handling																										All	Full run test with scoring
Phase 5: Documentation	TDP, Poster, Video, BOM completed																										All	Submission 3 weeks before competition
Phase 6: Competition	RoboCupJunior Rescue Maze 2026																										All	Competition day

Figure 1 : Gantt Chart

Sequence of Milestones

The sequence was planned based on build logic and technical dependency. Mechanical fabrication and motor control were completed first to establish a stable and moveable platform before any software development began. Once the robot could move reliably, navigation and mapping were developed on top of the motion foundation. Greek letter detection and cognitive target detection were developed simultaneously by different team members since both are fully independent pipelines with no shared dependency.

Testing Conditions per Iteration

Each iteration was tested under different lighting conditions and arena setups based on the competition rules. This included standard and dim lighting, various wall configurations, ramp angles up to 25 degrees, and coloured tile placements. Victim targets were placed at different positions and orientations to cover as many real competition scenarios as possible.

Impact of Past Performance on Subsequent Iterations

Each test result was reviewed and used to improve the next version. Poor motor alignment led to encoder-based synchronisation. Low letter detection accuracy triggered a new round of dataset collection with more varied images. Unreliable cognitive ring detection led to the template-anchoring approach. Navigation failures in early runs revealed premature tile registration, which was fixed by tightening the tile-entry confirmation logic.

Integration Plan

The robot is built around two processing layers working together to achieve full autonomous operation. The Jetson Orin Nano handles all high-level decisions including maze navigation, victim detection, and rescue kit deployment logic. The ESP32-S3 handles all real-time low-level tasks including motor control, colour sensing, servo actuation, and user input. Both processors communicate over USB serial at 115200 baud using a structured command and response protocol.

Figure 2 below shows the full system block diagram illustrating how all components are connected and how data flows between the two processors and their respective sensors and actuators.

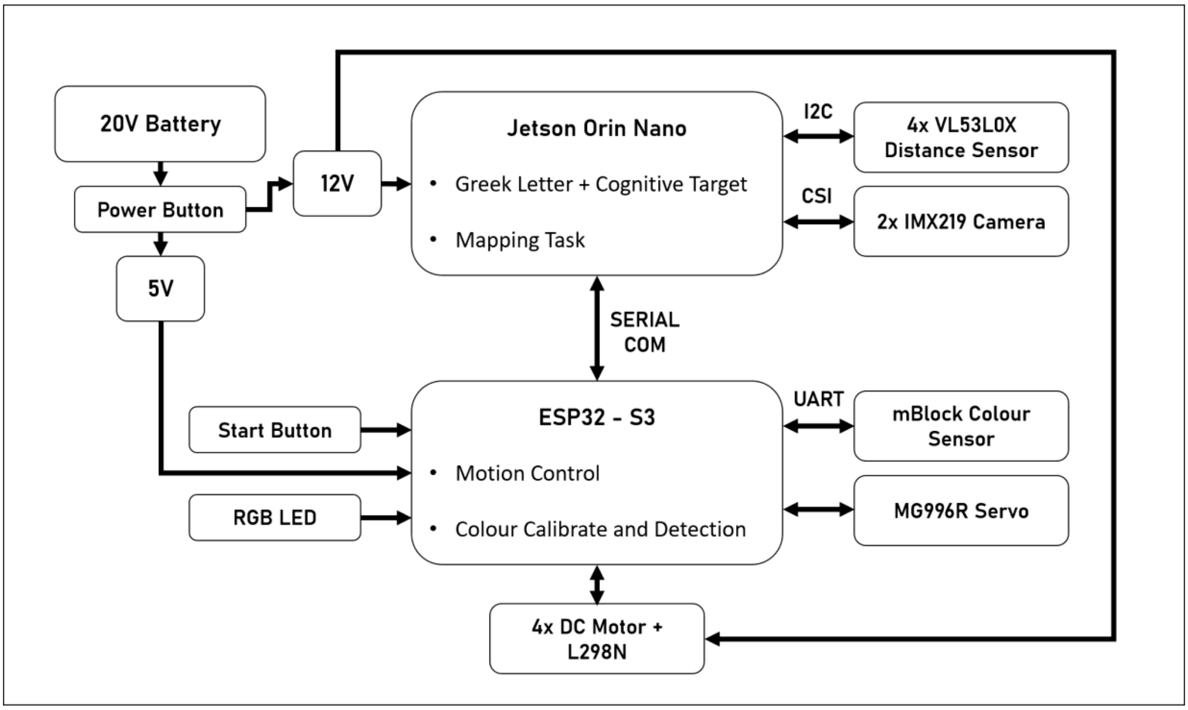


Figure 2 : PRIMEX FCoEE System Block Diagram

Serial Communication Protocol

- Commands sent from Jetson to ESP32: MOVE, TURN_LEFT, TURN_RIGHT, STOP, DROP_LEFT, DROP_RIGHT, BLINK_LED, CALIBRATE_COLOUR
- Responses sent from ESP32 to Jetson: COLOUR:<value>, ENCODER:<L,R>, BUTTON:PRESSED, DROP:DONE, STATUS:OK

Each requirement from the planning phase is satisfied by a specific hardware-software component pair:

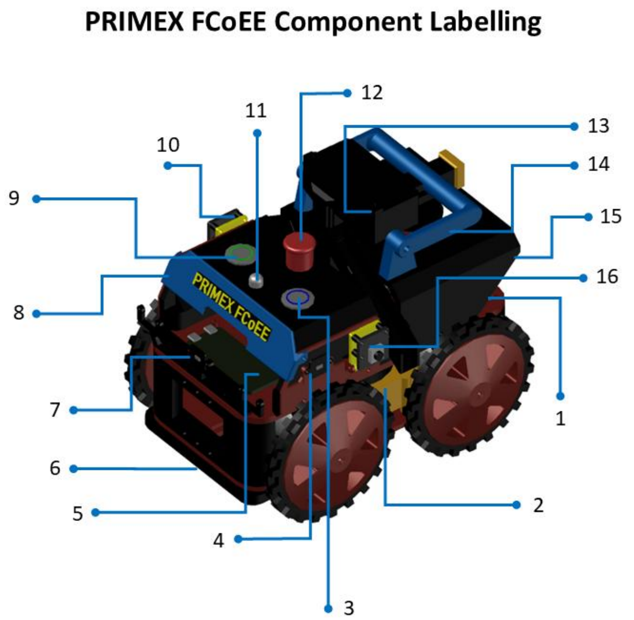
Requirement	Hardware Component	Software Module
Wall detection / navigation	4× VL53L0X ToF	Jetson: I2C polling, XSHUT multiplexing, distance threshold logic
Floor tile detection	mBot2 Colour Sensor	ESP32: UART packet read, calibration routine, reports to Jetson
Greek letter victim detection	2× IMX219 160° Camera (left + right)	Jetson: OpenCV pre-process, TensorFlow CNN inference
Cognitive target detection	2× IMX219 160° Camera (left + right)	Jetson: OpenCV contour detection, ring isolation, NumPy value sum

Motor drive and synchronisation	4× DC Motor + 2× L298N + encoders	ESP32: PWM control, encoder feedback, closed-loop sync
Rescue kit deployment	MG996R Servo + 3D printed deployer	ESP32: PWM servo, kick-left / kick-right on serial command
Start / calibration trigger	Momentary push button	ESP32: interrupt handler, reports BUTTON:PRESSED to Jetson
RGB LED victim signal	Dedicated RGB LED Indicator (GPIO 10)	ESP32: blink 500 ms ON / 500 ms OFF on BLINK_LED command
Checkpoint map save	Jetson internal storage (eMMC)	Jetson: writes 2D grid JSON on silver tile detection
Power management	20V battery, 12V reg, 5V reg	Hardware only: independent rails for motors/Jetson and ESP32/sensors

3. HARDWARE

Overview

PRIMEX FCoEE is a compact 4-wheel drive robot measuring 190 mm (W) × 230 mm (L) × 200 mm (H), well within the competition pathway width of 280 mm and the height limit of 250 mm. The entire chassis and body are custom 3D printed with no commercial robot kit frame used. Two independent processors handle distinct responsibilities: the Jetson Orin Nano manages all perception and high-level decision-making, while the ESP32-S3 handles real-time motor and actuator control. A 20V power tools battery provides the main power source, stepped down to 12V for motors and Jetson, and 5V for the ESP32 and low-power sensors. Figure 3 shows the full robot configuration with all 16 major components labelled.



No	Component
1	Jetson Orin Nano
2	20V 1.5Ah Battery
3	Power Button
4	SL Distance Sensor (VL53L0X)
5	ESP32 S3 Devkit Board
6	mBlock Colour Sensor
7	FC Distance Sensor (VL53L0X)
8	SR Distance Sensor (VL53L0X)
9	Start Button
10	Right Side IMX219 Camera
11	RGB LED Indicator
12	Stop Button
13	MG996R Servo + Drop Kit Mechanism
14	Main Handle
15	RC Distance Sensor (VL53L0X)
16	Left Side IMX219 Camera

Figure 3 : PRIMEX FCoEE Full Robot Configuration with Component Labelling

3.1 Mechanical Design and Manufacturing

Main Structure and Chassis

The chassis was designed from scratch using CAD software and printed entirely in-house. Three filament materials were selected for specific structural roles:

- ABS-CF (Carbon Fibre reinforced ABS): Used for the main body panels and upper structure. ABS-CF provides high stiffness and heat resistance, protecting electronics and surviving field impacts.
- PETG-CF (Carbon Fibre reinforced PETG): Used for the chassis base frame and motor mounts. PETG-CF offers excellent layer adhesion and toughness under torsional and bending loads from the drive system.
- PLA: Used for non-structural accessories such as sensor brackets, cable guides, camera mounts, and the main handle where weight saving is prioritised over structural demand.

This deliberate material selection is an innovative approach for a junior team. CF-reinforced materials reduce flex under load and improve positional accuracy of sensors mounted on the body, which directly benefits the consistency of ToF readings and camera alignment.

Actuators and Drivetrain

Four JGB37-520 12V DC motors with built-in encoders (76 RPM) drive the robot in a 4-wheel differential configuration. Two L298N H-bridge motor drivers, each controlling one left-right pair, allow independent speed and direction control. Encoder feedback from all four motors is read by the ESP32-S3, enabling closed-loop synchronisation that maintains straight-line movement with a measured lateral error of less than 1 mm.

Rescue Kit Deployment Mechanism

Rescue kits are 11×11×11 mm cubes printed in PLA (volume 1.331 cm^3, exceeding the 1 cm^3 minimum). The robot carries up to 8 kits in a magazine-style holder. An MG996R servo motor actuates the deployment arm to eject kits left or right depending on which wall the victim is detected on. The servo uses distinct angular positions for home, left kick, left release, right kick, and right release, allowing reliable repeatable deployment triggered by a serial command from Jetson after victim classification is confirmed. The mechanism implementing horizontal orientation with spring loaded from back. Slide ramp was set for both left and right side to ensure the kit can be drop on both sides. Figure 4 shows the detail of Rescue kit Deployment Mechanism

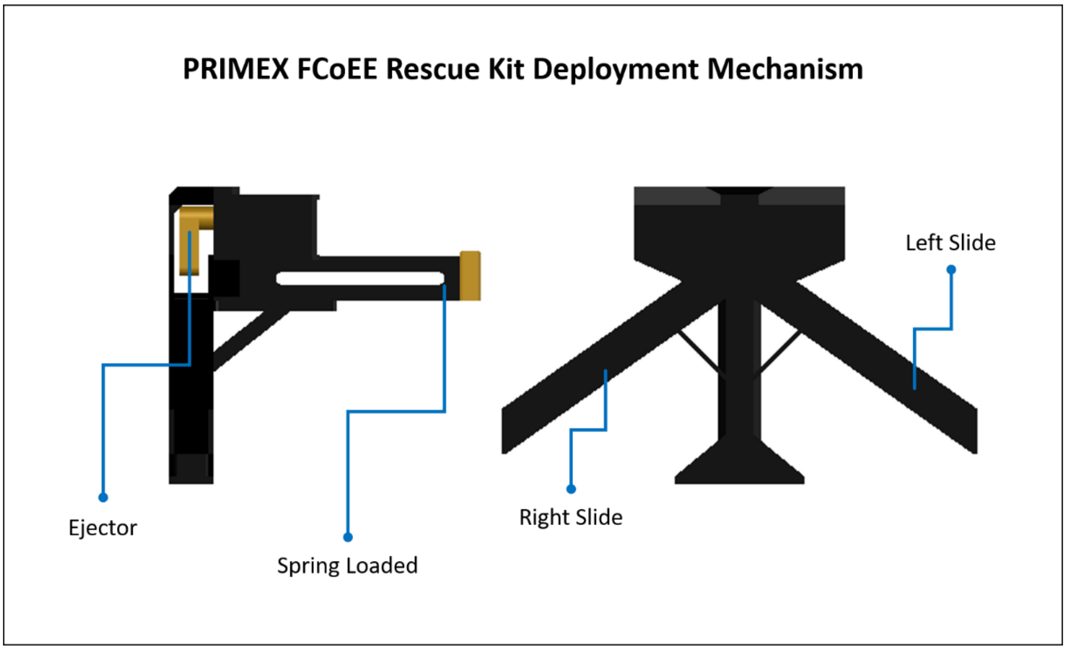


Figure 4 : PRIMEX FCoEE Rescue Kit Deployment Mechanism

Main Handle

A dedicated carry handle (component No. 14 in Figure 1) is 3D printed in PLA and mounted on the top of the robot body. The handle is positioned to allow a referee or team captain to pick up the robot safely during a Lack of Progress event without touching any sensor, camera, or electronic component. This satisfies the mandatory handle requirement under Rules §4.2 and ensures robot pick-up does not disturb the calibrated sensor positions.

Camera Mounting

Two IMX219 160° wide-angle camera modules (components No. 10 and No. 16 in Figure 1) are mounted symmetrically on the left and right sides of the robot body at approximately 100-110 mm height with a 15-20° downward angle. The wide 160° field of view allows victim detection on lateral walls while the robot is still moving forward, without requiring a stop-and-rotate sequence. Figure 5 shows the placement of manual tilting mechanism to give flexibility in adjustment during training and detection.

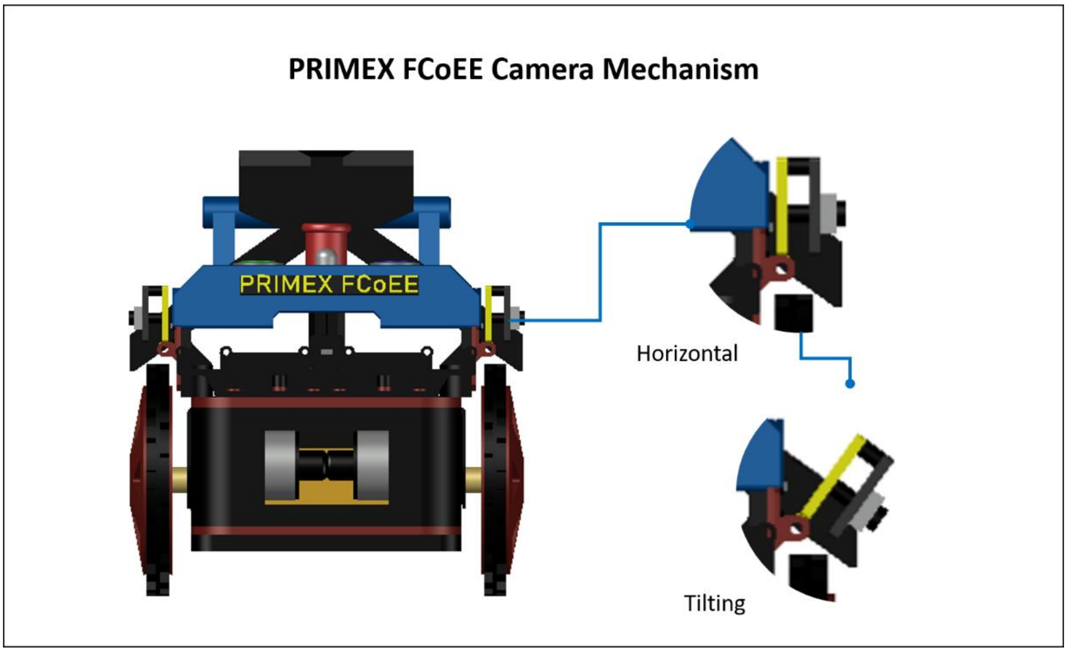


Figure 5 : PRIMEX FCoEE Camera Mechanism

Power Module

Swappable Battery mechanism using Power Tools Battery and docker, for faster battery swapping process.

Mechanical Testing

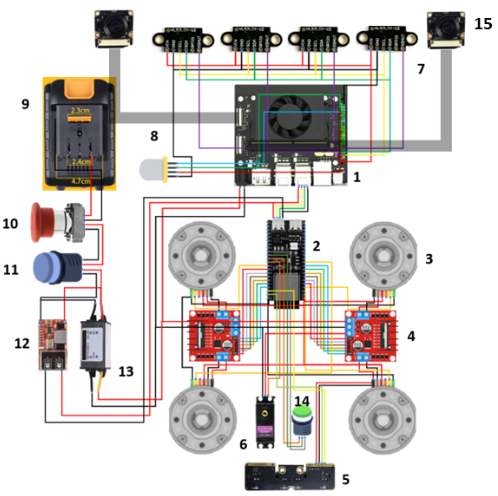
- Drop test: Robot dropped from 20 cm onto hard floor with no structural failure on chassis or mounts.
- Ramp traverse: Tested on 25° incline with all four motors maintaining traction and no chassis flex observed.
- Fit check: Robot width confirmed at 190 mm, clearing the 280 mm maze pathway by 45 mm on each side.
- Servo deployment: 50 consecutive deployments with no kit jam and consistent placement within 15 cm of simulated victim.
- Handle load test: Handle verified to support full robot weight (lift and carry) without deformation or detachment.

3.2 Electronic Design and Manufacturing

Main Controllers

Figure 6 shows the full electronic schematic with all 15 components labelled and their wiring connections.

PRIMEX FCoEE Electronic Schematic



No	Component
1	Jetson Orin Nano 8GB RAM and 512GB SSD
2	ESP32 S3 Devkit Board
3	JGB37-520 Hall Encoder DC Motor
4	L298N Motor Driver
5	mBlock Colour Sensor
6	MG996R Servo
7	VL53L0X Distance Sensor
8	RGB LED Indicator
9	20V 1.5Ah Battery Module
10	Stop Button
11	Power Button
12	5V USB Regulator
13	12V Power Regulator
14	Start Button
15	IMX219 Camera Module 160°

Figure 6 : PRIMEX FCoEE Electronic Schematic with Component Labelling

1. Jetson Orin Nano (High-Level Processor, component No. 1): Handles all vision processing, sensor fusion, navigation decisions, and serial communication to ESP32. Interfaces: CSI (cameras), I2C (ToF sensors), USB Serial (to ESP32).
2. ESP32-S3 DevKit (Low-Level Processor, component No. 2): Handles motor PWM, encoder synchronisation, servo control, colour sensor decoding, button interrupts, and RGB LED indicator. Interfaces: GPIO (PWM, encoder), UART (colour sensor), USB Serial (to Jetson).

Actuators

1. The drivetrain uses four JGB37-520 Hall Encoder DC Motors (component No. 3)
   a. Running at 76 RPM, arranged in a 4-wheel differential drive configuration. Each motor has a built-in quadrature Hall encoder that provides continuous position and speed feedback to the ESP32-S3. Two L298N dual H-bridge motor

drivers (component No. 4) are used, each controlling one left-right motor pair. The ESP32 sends PWM signals to the L298N inputs to set motor speed and direction independently for each side, enabling forward, reverse, and point turns.

2. Encoder pulses from all four motors
    a. Read via GPIO interrupts on the ESP32. The firmware counts pulses on each wheel and compares left and right totals continuously. If a deviation is detected, PWM values are adjusted in real time to correct the imbalance, achieving a straight-line lateral error of less than 1 mm over a 1-metre run.
3. The MG996R servo motor (component No. 6)
    a. Controls the rescue kit deployment arm. It receives PWM signals directly from the ESP32 and operates at five defined angular positions: home, left kick, left release, right kick, and right release. Deployment is triggered by a serial command from the Jetson following confirmed victim classification.

Sensors

1. VL53L0X Distance Sensor (4 units, component No. 7, Jetson):
    a. Wall distance measurement at Front-Centre, Side-Left, Side-Right, and Rear-Centre positions. Connected to Jetson via I2C with XSHUT address multiplexing to assign unique addresses on a shared bus. Pre-calibration error ±40 mm; post-calibration error ±1 mm.
2. IMX219 Camera Module 160° (2 units, component No. 15, Jetson):
    a. Left and right wall victim scanning. Connected to Jetson via CSI ribbon cable (22-way to 15-way FPC). Both cameras stream simultaneously for continuous bilateral victim detection.
3. mBlock Colour Sensor (1 unit, component No. 5, ESP32):
    a. Downward-facing floor tile colour detection (black/silver/blue/red). Connected to ESP32 via UART using mBlock packet protocol with manual per-session calibration before each scoring run.

Indicators and Controls

1. RGB LED Indicator (component No. 8, ESP32):
   a. Dedicated victim identification signal. Blinks at 500 ms ON / 500 ms OFF for 5 seconds during victim identification, and 1 s ON / 1 s OFF for 10 seconds when triggering the exit bonus at the start tile. The RGB capability also provides visual status feedback during calibration and system startup.
2. Start Button (component No. 14, ESP32):
   a. Momentary push button used to start the scoring run and trigger colour sensor calibration before each run, as required by Rules §4.2.
3. Power Button (component No. 11):
   a. Self-latch push button controlling the main 20V feed from the battery module.
4. Stop Button (component No. 10):
   a. Emergency stop button cutting the 12V motor rail without shutting down the Jetson, preserving map data in memory.

Power Subsystem

1. Main source: 20V 1.5Ah Battery Module (component No. 9).
2. 12V Power Regulator (component No. 13): Steps down 20V to 12V to power the 4 DC motors (No. 3) and the Jetson Orin Nano (No. 1).
3. 5V USB Regulator (component No. 12): Steps down 20V to 5V to power the ESP32-S3 (No. 2), VL53L0X sensors (No. 7), mBlock Colour Sensor (No. 5), MG996R Servo (No. 6), and RGB LED Indicator (No. 8).

Separate power rails isolate motor switching noise from sensitive sensor and processor circuits, improving measurement stability.

Electronic Testing

- L298N Motor Driver: PWM response verified at various duty cycles and correct rotation direction confirmed for all four motors before chassis assembly.
- Encoder Accuracy: All four encoders tested by rotating each wheel a fixed number of turns and comparing expected versus recorded pulse count. Consistent results confirmed across repeated tests.

- Motor Synchronisation: Encoder closed-loop control verified at less than 1 mm lateral deviation over a 1-metre straight run.
- Servo Angular Positions: MG996R verified at all five defined positions across 20 repeated cycles with no drift or overshoot observed.
- ToF Calibration: Error reduced from ±40 mm (uncalibrated) to ±1 mm (post-calibration) across all 4 sensors at 50-300 mm range.
- Colour Sensor: Manual calibration verified on all four tile colours under simulated venue lighting. Classification accuracy consistent after calibration.
- Serial Communication: All defined commands tested between Jetson and ESP32. No dropped packets or timeout over a 5-minute continuous stress test.
- Camera Initialisation: Both IMX219 cameras verified for simultaneous streaming on Jetson with no conflict between the two CSI ports.
- RGB LED: Blink timing verified at 500 ms ON / 500 ms OFF and exit bonus blink at 1 s ON / 1 s OFF using oscilloscope measurement.
- Power Rail Stability: 5V rail variation less than 0.1V under full load with all sensors, ESP32, servo, and RGB LED active simultaneously.

4. SOFTWARE

Overview

The software system is divided across two processors. The ESP32-S3 runs firmware written in C/C++ (Arduino framework) handling all time-critical low-level tasks. The Jetson Orin Nano runs Python 3 modules for perception, mapping, and decision-making. The two processors communicate through a structured command-response protocol over USB serial at 115200 baud. No pre-built OCR, line-following, or maze-solving libraries are used; all core algorithms are developed by the team.

4.1 Software Navigation Flow

Figure 7 illustrates the complete software execution flow of PRIMEX FCoEE from startup to end of run. The system begins with both the Jetson Orin Nano and ESP32-S3 in standby. The team captain first presses the calibrate button, which triggers an iterative calibration loop where the colour sensor calibrates one floor colour at a time until all four colours (black, silver, blue, and red) are successfully registered. Once calibration is complete, the 2D grid map is initialised and all four distance sensors (FC, SL, SR, RC) are read for the initial wall configuration. The team captain then presses the start button to begin the scoring run.

During navigation, the colour sensor continuously checks the floor tile beneath the robot. On a blue tile the robot stops for 5 seconds before continuing; on a black tile it stops immediately and reverses to avoid a Lack of Progress; on a silver tile the map is updated and saved to storage as a checkpoint. After tile handling, the system checks whether the maze is fully explored. If not, the robot applies autonomous motion using the Right-Hand Rule, choosing to move forward, turn left, or turn right based on wall readings. Both cameras simultaneously scan for victims during movement. When a victim is detected, it is classified, and the system determines whether the victim is harmed. If harmed, the robot drops the appropriate number of rescue kits before blinking the RGB LED. If not harmed, the LED blinks only. Once the maze is fully explored, the robot returns to the start tile to trigger the exit bonus and end the run.

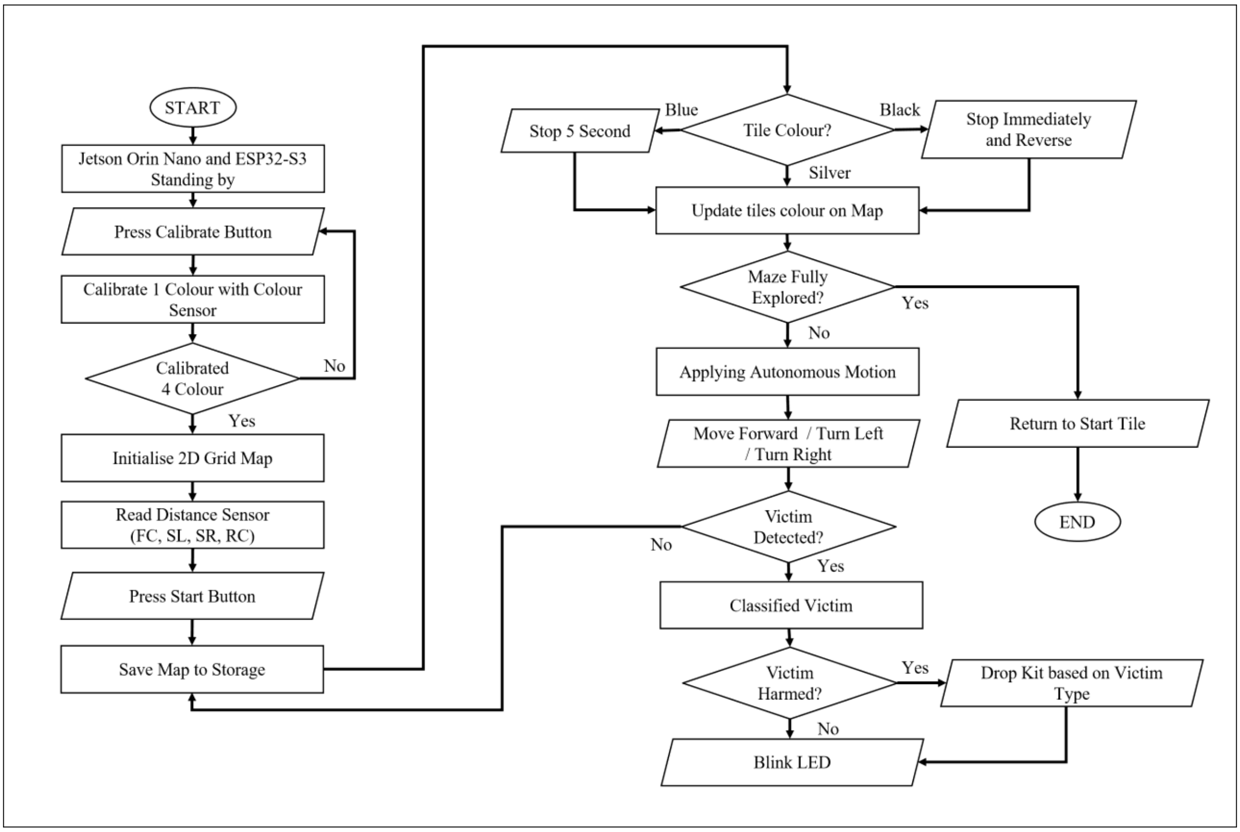


Figure 7 : PRIMEX FCoEE Operational Flowchart

4.2 General software architecture

Layered Architecture

1. Application Layer (Jetson): Main state machine cycling through EXPLORE, DETECT, DEPLOY, and RETURN states.
2. Perception Layer (Jetson): Camera vision pipelines (letter CNN and cognitive ring detection) and ToF sensor reading.
3. Navigation Layer (Jetson): Right-Hand Rule algorithm, 2D grid map management, tile type tracking, and LoP checkpoint recovery.
4. Communication Layer (Jetson/ESP32): Serial TX/RX command dispatcher handling all inter-processor messaging.
5. Actuation Layer (ESP32): Motor PWM with encoder synchronisation and servo deployment control.
6. Sensing Layer (ESP32): Colour sensor packet decoding, button interrupt handling, and RGB LED indicator control.

Navigation and Maze Mapping

The navigation module implements the Right-Hand Rule algorithm on a 2D grid map where each cell represents one 30×30 cm tile. Wall presence is inferred from the four ToF sensors at each tile position. The algorithm evaluates the right-hand wall first: if open it turns right, if blocked it continues forward, and if forward is also blocked it turns left or reverses. This guarantees full exploration of simply-connected maze sections without requiring a pre-built map.

The grid stores each cell state including visited status, wall configuration, and tile type. On detection of a silver checkpoint tile, the full map is serialised to JSON and written to Jetson storage. After a Lack of Progress restart, the saved map is reloaded so the robot resumes exploration with full knowledge of previously visited tiles, preventing redundant re-exploration.

Greek Letter Victim Detection

The detection pipeline is divided into three stages: dataset collection, model training, and live inference.

Stage 1 — Dataset Collection

- Images of the three letter victims (Φ, Ψ, and Ω) were manually captured by the team using the IMX219 cameras mounted on the robot. Captures were taken under varied conditions to build a robust dataset: multiple lighting environments (bright, dim, and mixed), rotation angles from 0° to 360°, and capture distances of 10 to 20 cm from the wall. Each letter was captured approximately 3,500 times, producing a total dataset of around 10,500 images across the three classes. Images were saved as grayscale at 28×28 pixels to match the model input size and reduce storage and processing load on the Jetson.

Stage 2 — Model Training

- The dataset was split 80/20 into training and validation sets. A lightweight TensorFlow Convolutional Neural Network (CNN) was trained with an image input size of 28×28 pixels, batch size of 32, and 20 training epochs. The model architecture was kept intentionally compact to enable fast inference on the Jetson Orin Nano without GPU bottleneck. Data augmentation including random rotation, brightness adjustment, and horizontal flipping was applied during training to further improve generalisation. The trained model was saved and deployed directly onto the Jetson for live on-device inference. The final model achieves 90% classification accuracy on the validation set.

Stage 3 — Live Inference and Deployment

- During a scoring run, both left and right camera streams are processed independently and simultaneously. Each frame is pre-processed with OpenCV: converted to grayscale, Gaussian-blurred, and thresholder to isolate the black letter region from the wall background. The cropped region is resized to 28×28 pixels and passed to the saved model for inference. If the confidence score reaches 85% or above, the classification is accepted. The robot then stops within 15 cm of the victim, triggers the RGB LED blink sequence at 500 ms ON / 500 ms OFF for 5 seconds, and sends either DROP_LEFT or DROP_RIGHT to the ESP32 based on which camera detected the victim.

Current model achieves 90% detection accuracy on the validation set.

Cognitive Target Detection

Cognitive targets consist of up to 5 concentric coloured rings whose sum determines the victim health status. The detection pipeline proceeds in four stages:

1. **Stage 1** (Outermost Circle Detection): OpenCV HoughCircles and contour analysis detect the largest circular region in the camera frame, establishing the target position and scale.
2. **Stage 2** (Ring Template Placement): Using the detected outer circle radius, a template of 5 concentric ring boundaries scaled to image pixels is overlaid on the detected circle.
3. **Stage 3** (Colour Sampling per Ring): Each ring region is sampled using NumPy array masking. The dominant colour (Black, Red, Yellow, Green, Blue) is classified using HSV threshold calibrated to competition lighting.
4. **Stage 4** (Health Status Computation): The colour-to-value mapping (Black = -2, Red = -1, Yellow = 0, Green = +1, Blue = +2) is applied to all 5 rings and summed. Sum 2 indicates Harmed (Φ) Sum 1 indicates Stable (Ψ), and Sum 0 indicates Unharmed (Ω). Values outside this range are rejected as fake targets.

Current performance: 80% ring layer detection accuracy and 90% per-ring colour classification accuracy.

4.3 Libraries and Packages Used

Jetson Orin Nano (Python 3)

1. OpenCV (cv2): Image pre-processing, contour detection, HoughCircles, HSV colour thresholding, and frame capture from both cameras
2. TensorFlow / Keras: CNN model training, saved model loading, and live inference for Greek letter classification
3. NumPy: Array masking for per-ring colour sampling in cognitive target detection and numerical computation
4. PySerial: USB serial communication with ESP32-S3 for sending commands and receiving sensor status
5. VL53L0X Python Library: I2C distance sensor reading with XSHUT multiplexing for all four ToF sensors
6. JSON (built-in): Serialising and saving the 2D grid map to storage at each checkpoint

ESP32-S3 (Arduino / C++)

1. Arduino Framework: Core firmware structure, GPIO handling, and interrupt service routines for encoder pulse counting
2. ESP32Servo: PWM servo control for MG996R rescue kit deployment mechanism
3. HardwareSerial: UART communication with mBot2 colour sensor using mBlock packet protocol

4.4 Innovative solutions

1. Engineering-Grade 3D Printed Chassis
    - The use of ABS-CF and PETG-CF materials provides a significantly higher stiffness-to-weight ratio than standard PLA, ensuring sensor brackets and camera mounts maintain their calibrated positions after impacts and ramp traversals. This directly improves the consistency of ToF distance readings and camera alignment feeding into both the navigation and vision algorithms.

2. Swappable Power Tool Battery System

- The robot uses a 20V 1.5Ah power tool battery mounted on a quick-release bracket, allowing the battery to be swapped in seconds without tools. This is a practical competition advantage when between scoring rounds, the team can replace a depleted battery with a fully charged one instantly, ensuring the robot always operates at full power capacity for every run. Standard fixed battery designs require the team to either wait for an onboard charge or rewire the power connection, both of which consume limited preparation time between rounds. The use of a standard power tool battery format also means replacement units are widely available and the battery management system is already built into the battery module itself, improving safety and reliability.

3. Dual-Camera Lateral Scanning
    - Cameras are mounted on the left and right sides rather than front-facing, allowing the robot to continuously scan both walls for victims while moving forward. The 160° wide-angle IMX219 modules ensure full wall coverage even at close range, and the adjustable mounting angle allows field-day tuning based on arena lighting conditions. This eliminates the stop-and-rotate overhead that a front-facing camera approach would require.

4. Right-Hand Rule with Persistent Grid Map and Checkpoint Recovery
    - The navigation algorithm maintains a persistent 2D grid map saved to non-volatile storage at every checkpoint. After a Lack of Progress event, the saved map is reloaded at the restart position, allowing the robot to resume exploration knowing which tiles were already visited and avoiding redundant re-traversal of known areas.

5. Custom-Trained Lightweight CNN for On-Device Greek Letter Detection
    - Rather than using a general-purpose OCR library or pre-trained model, the team built and trained a custom Convolutional Neural Network specifically for the three competition letter victims (Φ, Ψ, Ω). The dataset of approximately 10,500 images was manually captured using the robot's own cameras under real competition-like conditions, covering varied lighting, rotation angles from 0° to 360°, and distances of 10 to 20 cm. The model was designed to be intentionally lightweight with a 28×28-pixel input size, enabling fast on-device inference on

the Jetson Orin Nano without requiring a dedicated GPU pipeline. This custom approach outperforms generic OCR solutions for this specific task because the model is trained exclusively on the target characters under the exact visual conditions the robot will encounter, achieving 90% classification accuracy on the validation set.

6. Concentric Ring Template Strategy for Cognitive Target Detection
   - Rather than detecting each ring independently through edge detection, which is fragile under varying contrast and lighting, the team developed a template-anchored approach: detect the outermost circle first (most reliable as largest contour), then mathematically derive all inner ring positions from the known 1:2:3:4:5 diameter ratio. This reduces the problem from five independent detection tasks to one detection plus four computed lookups, significantly improving robustness. This approach was developed entirely by P4 without reference to existing cognitive target detection implementations.

5. PERFORMANCE EVALUATION

5.1 Testing Methodology

Performance evaluation was conducted through iterative field tests on a custom-built practice arena replicating competition tile dimensions (30×30 cm) and wall configurations. Each subsystem was tested independently first (unit testing), then together as an integrated system (integration testing). Results were recorded and used to guide subsequent development iterations.

Subsystem Performance Results

Subsystem	Metric	Pre-Improvement	Post-Improvement	Method
ToF Distance Sensors	Measurement error	±40 mm (uncalibrated)	±1 mm (per-sensor offset calibration)	Ruler at 10, 20, 30 cm
Motor Synchronisation	Lateral deviation over 1 m	~5 mm (open loop)	< 1 mm (encoder closed-loop)	Tape measure on marked floor
Greek Letter Detection	Classification accuracy	~65% (initial small dataset)	90% (augmented dataset, tuned CNN)	3500 validation images per class
Cognitive Ring Detection	Ring layer detection accuracy	~55% (contour only)	80% (template anchoring method)	100 test images per target type
Cognitive Ring Detection	Per-ring colour accuracy	~70% (RGB thresholding)	90% (HSV thresholding + calibration)	Manual verification on 50 targets
Maze Navigation	Successful traversal with correct action	~35% (early algorithm)	60% (Right-Hand Rule + grid map)	Full arena traversal runs (10 runs)

5.2 Development and Detection Result

Robot Development

Figure 8 shows the final PRIMEX FCoEE robot after the January–June 2026 development cycle. The robot features an ABS-CF body, PETG-CF chassis, four-wheel differential drive system, dual IMX219 cameras, a spring-loaded rescue kit mechanism, and a carry handle. The top panel includes Start, Stop, and Power buttons along with an RGB LED indicator, demonstrating compliance with the physical requirements specified in Rule 4.2.

Figure 9 presents the robot from three different angles, highlighting the internal electronics layout, dual IMX219 cameras, Jetson Orin Nano interface ports, battery compartment, wheel configuration, carry handle, and the robot's compact chassis design.

Figure 8 : Full Developed PRIMEX FCoEE Robot

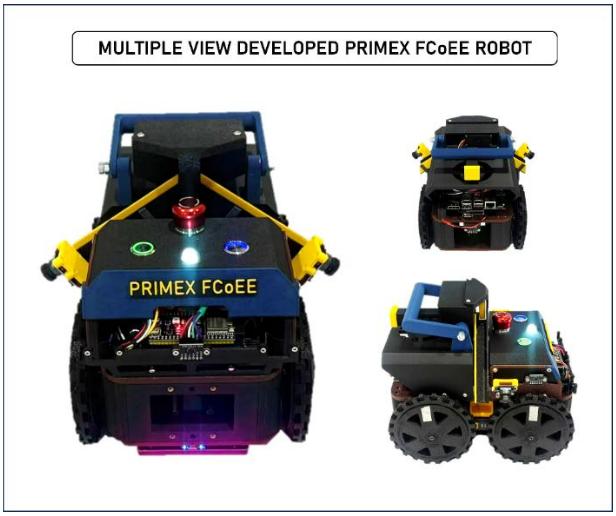


Figure 9 : Multiple View PRIMEX FCoEE Robot

Greek Letter Detection Result

Figure 10 shows real-time letter detection results from the left and right IMX219 cameras during a scoring run test. Both cameras accurately classify Omega (Ω), Psi (Ψ), and Phi (Φ) with confidence scores above 99%, demonstrating the reliability and consistency of the custom-trained TensorFlow CNN across different viewing angles. The detected class triggers the corresponding LED blink sequence and rescue kit deployment command.

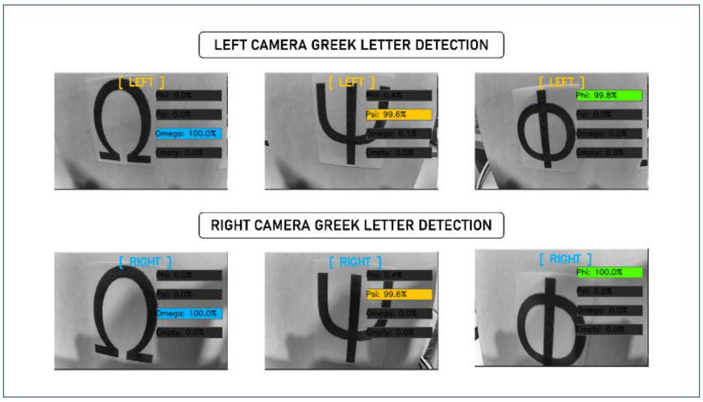


Figure 10 : Greek Letter Detection Result

Cognitive Target Detection

Figure 11 shows real-time cognitive target detection results from both cameras, with ring boundaries and colour classifications overlaid on the camera feed. The system accurately identifies the five ring colours across multiple target configurations, demonstrating reliable target localisation and HSV-based colour classification. The results validate the ring detection pipeline and confirm accurate colour recognition before health status calculation.

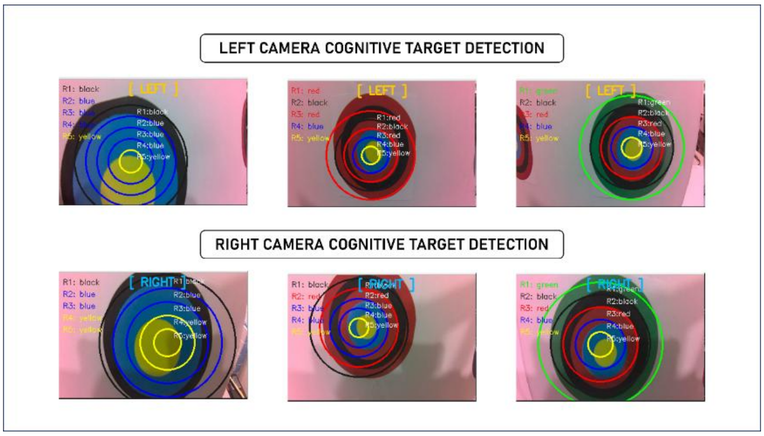


Figure 11 : Cognitive Target Detection Result

5.3 Impact on Development

Each test result directly drove a specific code or design change. The ToF calibration error led to a per-sensor offset correction table stored in Jetson memory. The initial low letter detection accuracy triggered a second dataset collection round under varied lighting and rotation, increasing training data from approximately 500 to 3500 images per class. The cognitive ring improvement came from replacing independent contour detection per ring with the template-anchoring strategy. Navigation improvement came from refining the wall detection threshold and adding a tile-entry confirmation step requiring the robot to be more than 50% inside a tile before recording it as visited, matching the competition rules definition.

5.4 Known Limitations and Ongoing Work

- Maze mapping success at 60% remains the current development focus. Edge cases involving dead ends and U-turns in the Right-Hand Rule algorithm are being refined.
- Cognitive target colour detection accuracy reduces under very low lighting (below 50 lux). Adaptive HSV threshold adjustment based on ambient brightness is being investigated.
- The current ToF distance sensing uses I2C address multiplexing via XSHUT pins across four sensors on a shared bus. Rapid sequential address switching between sensors occasionally causes brief flip-flop readings where a sensor momentarily returns an incorrect distance value. The team is investigating two potential solutions: rewiring each sensor to a dedicated I2C bus, or replacing the VL53L0X modules with sensors that support native multi-sensor operation without address conflicts.

6. Conclusion

PRIMEX FCoEE has developed a fully autonomous RoboCupJunior Rescue Maze robot that demonstrates a high level of technical integration across mechanical design, embedded systems, and computer vision. The robot successfully demonstrates five core technical contributions. The foundation begins with high-performance hardware built on engineering-grade CF-reinforced 3D-printed materials and a dual-processor architecture that separates high-level vision and navigation from real-time low-level control. Built on top of this platform, a fully autonomous maze navigation system using Right-Hand Rule and real-time 2D grid mapping enables the robot to explore unknown environments with zero pre-mapping and zero hardcoded logic. For victim identification, a custom-trained lightweight CNN achieves 90% classification accuracy on Greek letter victims, while a template-based cognitive target detection pipeline achieves 80% ring layer detection and 90% per-ring colour classification. Underpinning all of this is an adaptive floor colour sensing system with per-session local calibration that ensures reliable tile classification under varying venue lighting conditions. Maze navigation currently achieves 60% successful tile coverage with correct action, which remains the primary area of active development alongside improvements to distance sensing reliability and cognitive target detection under low lighting.

The team has grown significantly through this project, from initial component selection and rule analysis in January 2026 to a fully integrated, competition-ready robot by June 2026. The skills developed in embedded firmware, deep learning model training, computer vision pipeline design, and system integration provide a strong foundation for future participation. PRIMEX FCoEE is committed to continuing improvement and to openly sharing the team's developments with the RoboCupJunior community.